Cold Referee Report: Decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ Concept Satisfiability

Packet 3 of 3: Verified Steps, Minor Findings, Probe Log, Computational Appendix, Recommendations

Eighth review of the split-forest stack

July 9, 2026

A. Steps checked and found sound (S1–S10)

Each item states what was checked and how (*computed* = fresh machine verification, Section D; *reconstructed* = full proof re-derivation by this reviewer; *read-through* = checked against the text with spot checks).

S1. The RCC5 composition table (computed). Derived independently from set semantics over all 63 nonempty regions of a 6-element universe, all 63^3 triples; exact match with the base Section 2 table in all 25 cells, including the two asymmetric EQ-bearing cells $\mathrm{Comp}(\mathsf{PP}, \mathsf{PPI}) = \mathrm{Rel}$ and $\mathrm{Comp}(\mathsf{PPI}, \mathsf{PP}) = \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}\}$, a classic transcription-error site. Taken as ground truth thereafter.

S2. Facts 2.1–2.4 (computed). Horizontal absorption, vertical transitivity, no horizontal dead ends, and EQ-never-forced (every EQ-bearing cell other than $\mathrm{Comp}(\mathsf{EQ}, \mathsf{EQ})$ also contains PP, PPI, PO) all hold.

S3. The AV semantic layer: Lemmas 3.1, 3.2, Theorem 3.3 (reconstructed). Lemma 3.1 is immediate. Lemma 3.2: the patched family $(B_t, N_t^* \restriction B_t)_{t \in Q}$ is tree-shaped — universal sources are active throughout Q; a live occurrence's manifestation set within Q is an intersection of two subtrees of a tree, hence connected — and adjacent restrictions agree by AV3, so the finite patchwork theorem applies; for occurrences of F active nowhere in Q (a case the statement covers but the proof ignores; F4 below) the completion extends by DR-isolation, verified by computation. Theorem 3.3: with C_F read as "$\lambda \restriction F$ extends to some closed atomic network on F_{Q_F} extending all declared data" (F5 below), each C_F is clopen; the finite-intersection property follows by taking a connected $Q \supseteq \bigcup_i Q_{F_i}$ and one Lemma-3.2 completion, whose restrictions witness membership in every C_{F_i}; a point λ^* of the intersection satisfies (R1)–(R3) because every 3-element restriction of λ^* extends to a closed network and restrictions of closed networks are closed; λ^* extends every declared live and active value because for each declared pair one may take F to be that pair and Q_F a bag where it is co-active. **The realization direction is, at the semantic level, genuinely repaired**; this is the round's real achievement.

S4. Coverage and truth: Lemma 4.1 and Theorem 4.3 (reconstructed). Two points the text uses but does not isolate were verified. (i) *Global rigidity for source–source pairs*: two universal sources are both active at *every* bag, so AV3 agreement across every adjacency propagates a single value tree-wide; hence Lemma 4.1's "any bag with a live manifestation" is well-defined. (ii) *Bidirectional coverage*: AV4 checks pairs (universal source, live port) in the source-to-port direction only, yet the truth lemma needs both directions for pairs never co-bagged.

This closes because every occurrence is live somewhere: for $\forall R.D \in \mathrm{tp}(x)$ with target y, apply AV4 at a bag where y is live (x virtually active there); for $\forall R.D \in \mathrm{tp}(p)$ with p live and a never-co-bagged universal target x, note $p \in \mathrm{Univ}_T$ too, and apply AV4 at a bag where x is live, with p as the virtual source. Both directions land on live-port instances of AV4; combined with rigidity, the truth lemma's universal case is sound in both the positive and the contrapositive (type-completeness) reading. The existential case is sound given AV5's inherited witness machinery, modulo F2.

S5. Extraction, Theorem 5.1, at the set level (reconstructed). Defining N_t^* by true relations of the underlying model makes AV1–AV3 restrictions of one global relation (trivially coherent, nested-compatible), AV4 semantic, AV6 vacuous. Sound modulo the width provenance and D4 accounting (F2, F3).

S6. Strong equality end to end (reconstructed; small cases computed). AV1(d) and AV2 forbid declared EQ off the diagonal; completions are atomic by construction; Fact 2.4 (computed) guarantees non-EQ choices are never forced through any triangle; compactness preserves this since every 2-element restriction of λ^* extends to an atomic completion. Blocking and repetition remain distinct occurrence classes (round-12 Appendices R, G, retained). Probes p4/p5 below exercise the sharpest case — semantic uniqueness without syntactic identity — and the machinery handles it correctly in both directions.

S7. The parity layer (read-through; conditional on F1). Per-request odd priorities with fulfilment switching to even make any infinitely deferred request an odd-dominated branch, hence rejecting; this per-copy design correctly repairs the round-12 round-robin defect (F6(i) of the earlier reviews). It lives inside the unproved automaton, so its soundness is conditional.

S8. Withdrawal audit (reconstructed). Everything round 14 withdraws from round 13 is discharged at the semantic level: D1 by AV2's exact rows; D2's protective function by AV4; D5 by AV3 rigidity; the Λ defaults by AV6 plus compactness (non-universal sources impose no universal checks, so undeclared pairs are genuinely free); the round-13 coverage lemma by Lemma 4.1; D3 correctly demoted (nothing load-bearing cites it). The two exceptions are the automaton (F1) and Section 8 (F3). One silent dependency break found: F6 below.

S9. Patchwork, Theorem 2.5 (computed in the small; external in general). Exhaustively verified for the two smallest amalgamation shapes: all 427 agreeing pairs of closed atomic 3-networks over a 2-overlap and all 21,094 agreeing pairs of closed atomic 4-networks over a 3-overlap admit closed completions (0 failures). The general statement remains an external input, correctly flagged as such by the base manuscript; the stack's reliance on it should stay prominent.

S10. Corollary 5.2, G2a absorption (read-through; arithmetic computed). Under AV the tower's ancestors are virtual: live width 2 with $m-1$ virtual shadows, versus the withdrawn fallback's $m+1$ live ports. Row values consistent with the table ($\rho(a_k, a_0) = \mathsf{PPI}$ by iterated $\mathrm{Comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$). The G2a family is genuinely absorbed; note that the same table row is what makes the annotation route impossible in F1.2.

B. Minor findings (F4–F7)

F4 (minor). Lemma 3.2 statement/use mismatch. The statement admits finite F containing occurrences active nowhere in Q; the proof's patchwork covers only $\bigcup_{t \in Q} B_t$. Verified repair (computed): a closed atomic network extends over fresh points related DR to everything, since $\mathsf{DR} \in \mathrm{Comp}(X, \mathsf{DR})$ and $\mathsf{DR} \in \mathrm{Comp}(\mathsf{DR}, X)$ for every atomic X, and $\mathrm{Comp}(\mathsf{DR}, \mathsf{DR}) = \mathrm{Rel}$; 0 failures over all closed networks with $n \leq 4$. Alternatively, restate the lemma with the covering hypothesis Theorem 3.3 already enforces.

F5 (minor). Theorem 3.3's C_F is under-specified. "Extendable to one of the finite completions above" does not fix the family of completions or the subtree; as written, neither closedness of C_F nor the extension property of λ^* strictly follows. Verified repair: fix Q_F per F (connected, containing a live manifestation of each non-universal member of F) and define a completion as *any* closed atomic network on F_{Q_F} extending all declared live and active data there; the FIP and limit arguments then go through as in S3. Wording only, but this is the theorem the whole round rests on; it should be exact.

F6 (minor; feeds F1). Broken citation chain into round-12 Appendix H.2. H.2's shadow states update relation vectors "declared in the adjacent bag label." The effective stack no longer declares any such thing (round 14: shadows "are not listed ... in a bag label"), so every surviving appeal to round-12 automaton mechanics inherits a dangling reference; and Section 6's claim that the alphabet is "no larger than round 12" is vacuously safe but misleading — the round-12 alphabet's row-annotation component has been deleted, not preserved. A consolidated statement of the effective validity clauses and label format is needed (R5).

F7 (minor). Reproducibility. WP14–WP25 are cited with pasted outputs but no scripts are included in the stack. The finite claims I depended on were re-verified independently here (Section D); the remaining WP claims (e.g., the three-bag round-12 counterexample WP15) were treated as historical context only. Scripts should ship with the manuscript (R6).

C. Adversarial probes attempted (p1–p7)

p1 Table and facts: computed; all pass (S1, S2). **p2** WP20 triangle $x\,\mathsf{PP}\,y$, $y\,\mathsf{DR}\,p$, $x\,\mathsf{PO}\,p$: computed; correctly rejected, $\mathrm{Comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$. **p3** G2a tower $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top \sqcap \forall\mathsf{DR}.A$: satisfiable (nested chain; no DR pairs, so $\forall\mathsf{DR}.A$ is vacuous — computed sanity check); width absorption verified (S10). **p4** Shared-witness uniqueness tower (Packet 2, F2): a satisfiable concept forcing one globally unique A-element to witness $\exists\mathsf{PPI}.A$ at every element of an infinite tower; the abstraction must reuse a single occurrence with a connected trace (+1 width) rather than drag sources — absorbed by the machinery, but only under a presentation choice no lemma legislates; reinforces F2/F3. **p5** EQ-forcing: attempted to force EQ between distinct occurrences via surrounding declared values; blocked by Fact 2.4 plus built-in atomicity (computed in the small); in the realization direction, a hypothetical abstraction with two distinct "unique-A" occurrences is correctly killed by AV4 at a live bag (each is a universal source forcing $\neg A$ on the other under every off-diagonal relation). Strong equality survives its sharpest test in both directions. **p6** Infinite deferral: per-copy odd priority rejects; no accepting run can defer a request forever (conditional on F1). **p7** Accept-an-unsatisfiable-concept: I could *not* construct a concrete unsatisfiable concept that the procedure accepts, for the decisive reason that Section 6 pins down no procedure to run; the two-port configuration of Packet 2 (F1.3) is the closest constructible object — it defeats every check the sketch describes and is machine-checked at the composition level. Likewise no satisfiable concept exceeding width K_A was found once Corollary 5.2 is granted; F2/F3 stand as unproved obligations, not exhibited failures.

D. Computational appendix

Environment: Python 3, single fresh script (`verify_rcc5.py`, available on request), no dependencies. Regions are nonempty subsets of $\{0,\ldots,5\}$; $\mathrm{rel}(X,Y)$ classifies by $=, \subsetneq, \supsetneq$, overlap, disjoint; composition is derived existentially over all region triples; networks are complete atomic maps with converse and diagonal EQ; closure checks all ordered triples against the derived table; completions are exhaustive over the four off-diagonal atoms per unknown pair. Key outputs, verbatim:

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1. Composition table derived from set semantics (|U|=6, 63 regions, 63^3 triples):
   match with manuscript Section 2 table: EXACT
2. Fact 2.1 (horizontal absorption): True
   Fact 2.2 (vertical transitivity): True
   Fact 2.3 (no horizontal dead ends): True
   Fact 2.4 (EQ never forced) on cells [('PP','PPI'),('PPI','PP'),
   ('PO','PO'),('DR','DR')] : True
3. WP20 triangle x PP y, y DR p, x PO p:  Comp(PP,DR) = ['DR'] -> PO admitted: False
4. Closed complete atomic RCC5 networks: n=3: 41   n=4: 916
5a. Patchwork 3+3 over 2-overlap: 427 agreeing pairs, 0 amalgamation failures
5b. Patchwork 4+4 over 3-overlap: 21094 agreeing pairs, 0 amalgamation failures
6. DR-isolation (fresh point DR to all) preserves closure: n=3 failures: 0
   n=4 failures: 0
7. F1 witness: rows row(x,p)=PP, row(y,p)=PPI, row(x,q)=PP, row(y,q)=DR
   e forced via p: Comp(PP,PP) = ['PP']
   e forced via q: Comp(PP,DR) = ['DR']
   intersection: [] (empty => no coherent N* exists)
   source-triangle (PP,PP,DR): DR in Comp(PP,PP)? False
9. G2a model sanity: pairwise relations on nested chain: ['PP','PPI']
   (no DR pairs; forall DR.A vacuous; concept satisfiable)
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The derived table (rows compose columns; Rel abbreviates all five atoms):

$\circ$	EQ	PP	PPI	PO	DR
EQ	EQ	PP	PPI	PO	DR
PP	PP	PP	Rel	PP, PO, DR	DR
PPI	PPI	EQ, PP, PPI, PO	PPI	PPI, PO	PPI, PO, DR
PO	PO	PP, PO	PPI, PO, DR	Rel	PPI, PO, DR
DR	DR	PP, PO, DR	DR	PP, PO, DR	Rel

E. Recommendation and requests for round 15

Recommendation: major revision (gap). The semantic half of the programme is now in genuinely good shape — Theorem 3.3 and Theorem 4.3 are the strongest versions of these results in the stack's history, and the strong-equality treatment is correct end to end. The decision half is not written. Concretely:

R1. State Theorem C for the AV language as an existential-enrichment projection (Packet 2, F1.1), give the complete spawning discipline for single, pair, and triple monitors, and prove *soundness*: an accepting run induces a single coherent N^*. The two-port configuration of F1.3 is the acceptance-without-validity test any such design must fail-safe against.

R2. Either prove a bounded-relevance or bounded-annotation lemma making the enrichment tape-representable (with a proven congruence if sources are coalesced), or replace the tree-automaton layer by a mosaic / type-elimination argument executed on the infinitary AV abstraction directly; Theorems 3.3/4.3 already supply the model-construction half of such an argument.

R3. Prove the request-delay/width lemma: a bounded-delay normal form for vertical requests (or a witness-trace-extension policy as in probe p4), plus an explicit K_A accounting term for simultaneously live request sources (Packet 2, F2).

R4. Issue an erratum resolving the round-13 Section 8 scope contradiction, and re-prove Lemma 8.2 with a single explicitly decreasing measure covering (g2)–(g5) (Packet 2, F3).

R5. Publish one consolidated document stating the effective validity clauses, label format, and alphabet — the three-layer delta structure has already produced one silent dependency break (F6) and one scope contradiction (F3a).

R6. Ship the WP scripts (F7).

End of Packet 3 and of the report. Verdict, summary, and confidence: Packet 1. Major findings with witnesses: Packet 2.